Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/14/2016
PI Name:	Rickus, Jenna Ph.D.		
Project Title:	Sporesat 2 Mission: Investigating Biophysical Mechanisms of Plant Gravisensing Using a Lab-On-A-Chip Approach (CoPI=Rickus)		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture (2) Bioregenerative Life Support		
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Zip Code:	47907	**Congressional District:**	4
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 SALMON--Astrobiology & FSB NNH08ZDA009O-SCMAFSB
Start Date:	11/01/2014	**End Date:**	09/30/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	Small Satellites		
Flight Assignment:	NOTE: Extended to 9/30/2016 per NSSC information (Ed., 6/16/16) NOTE: Extended to 5/31/2016 (original end date 9/30/2015 and then extended to 12/31/2015), per NSSC information (Ed.,1/11/16) NOTE: Extended to 12/31/2015 (original end date 9/30/2015), per NSSC information (Ed., 8/19/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roux, Stanley Ph.D. (Co-PI: University of Texas, Austin)		
Grant/Contract No.:	NNX15AB86A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Ed. NOTE (June 2015): Continuation of "Sporesat Mission: Investigating biophysical mechanisms of plant gravisensing using a lab-on-a-chip approach"--grant NNX11AF48A for SporeSat 2. SporeSat is a Small Complete Mission based on a Mission of Opportunity-2 nanosatellite designed for real time measurement of calcium signaling and maintenance of variable gravity treatments during flight. The impetus for SporeSat is to gain a deeper knowledge of the molecular and biophysical mechanisms for plant gravity sensing by studying the activation of cell gravity-sensing and electrophysiological signaling in the single-cell model system, Ceratopteris richardii, using a novel 'lab-on-a-chip' microsensor technology platform called a CEL-C advanced bioCD. The science and technical objectives as outlined in the original SporeSat proposal: Science Objectives: 1) Determine gravity thresholds for Ca2+ channel activation in wild-type spores; 2) Observe Ca2+ current response of gene knockout mutants. Technology Objectives: 1) Design and fabricate the CEL-C advanced bioCD; 2) Integrate bioCD with data acquisition hardware and ground-based validation experiments; 3) Develop and integrate centrifuge hardware; 4) Operate bioCD in space-relevant environment; 5) Conduct spaceflight experiments. In 2013, SporeSat was adapted to a 2 flight mission strategy with flight 1 focused on the technology objectives and flight 2 focused on the scientific objectives. In April 2014, SporeSat-1 was deployed from a SpaceX Falcon 9 launched from Cape Canaveral as part of CRS-3. SporeSat-1 was successfully commanded and science data downloaded. However, the OLED light source failed and one of the three bioCDs overheated (40C) above its target set point (29C). These issues were addressed by the Ames engineering team via a Failure Review Board process. SporeSat1 also faced other technical challenges related to the biology. At the time of SporeSat1, the strain of spore that had been used for all prior gravity modulated calcium measurements, Hn-n, was not physically available anywhere in the world. Instead the next best strain, Rn5, was used but was limited by a lack of prior calcium data as well as reduced germination rates due to age of spore batch (~5 years). In addition, the available methods of generating the desired caclium mutants were not reliable. Since that time, Co-Principal Investigator Stan Roux and Mari Salmi produced new batches of Hn-n spores and new methods of genetic manipulation of spores became available. Among the improvements made prior to SporeSat2 was an improved bioCD enclosed assembly (EA) wall design. The first SporeSat EA wall design contained an agar volume that was determined to be insufficient for the fern spores during storage. The EA walls and spore wells were redesigned to increase the agar volume available for each spore and was used in SporeSat-1 flight and ground controls. The new design, while an improvement for the spores, created an unintentional electrical coupling of the Ca2+ sensors. In addition this change in wall design may have contributed to the moisture leakage that was involved in OLED and temperature control failure. We therefore created a design upgrade for SporeSat-2 that retained a higher agar volume but that electrically isolated the spore wells and Ca2+ electrode pairs. In November of 2014 SporeSat2 began and was parsed into a Phase I (ground) and Phase II (flight). The primary goal of Phase I was to demonstrate that the Hn-n and/or Rn5 fern spores exhibit a reliable and repeatable Ca2+ current that is measureable by the bioCD during germination under ideal ground conditions. In this report we focus on the major accomplishments achieved during the Jan 2015 – May 2016 period which encompassed SporeSat2 Phase 1. The SporeSat-2 experimental paradigm and geometry is different from any previous experimental set ups. As in any biosensor experiment, the Ca2+ ISEs are sampling at a particular location in space and time, a dynamic calcium current that exists across the 3D surface of the fern spore. To assist in development of the biological model based on sensor recordings, we have developed a 3D simulation model that predicts the 3D calcium concentration profiles in the SporeSat wells as well as the sensor recordings over time based on the biological flux from the fern spore. We will therefore be able to compare our experimental data with our predictions for different conditions (angle off gravity, under rotation) as well as for different calcium mutants.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Astronaut health in long term space missions is likely to be dependent on and benefit from plants. On Earth we depend on plants for food, clean air, and waste treatment. The same will be true in space. If we as humans are to achieve the kind of space exploration and habitation that we all imagine and dream of, we are going to need plants up there with us. Yet on the International Space Station (ISS), gravity is less than one one-thousandth of the gravity here on Earth. On the Moon, it is about 17% of Earth's gravity. In addition, there are astonishing similarities in signaling within cells across species. As in plants, calcium plays important roles in human cell development, function, and dysfunction. Understanding gravity sensing in plant cells is likely to give clues to gravity sensing in other organisms including humans. In addition, the technology that we developed for these experiments establishes a platform to ask other questions regarding human cell responses on Earth and in space. We are already looking at how we can adapt the technology to study pancreatic beta cells and diabetes.
Task Progress:	• Production of 18 bioCDs, which were used for SporeSat-2 Phase 1 (in addition to the 7 bioCDs that carried over from SporeSat1). • 50% completion of another 18 bioCDs (currently at step 10 of 17 in process). • Completion of a 3D finite element simulation model that predicted a bioCD response of 5 – 20 mV based on the working biological hypothesis. The small size of the expected signal highlighted the need to reduce the variation in negative controls and increase the signal magnitude in order to reliably resolve the spore signal with statistical significance. • Built and tested calcium optrode for independent validation of calcium signals. • A significant reduction in negative control and baseline variability by improving o Spore loading consistency and quality via implementation of automated loading at Ames ; o Elucidation of dryout and incomplete rehydration of the ion selective electrode during spore loading as a root cause of railed wells; o Overall improvement in quality and documentation of well and ion selective electrode conditions during spore and agar loading; o Amplifier tuning: reduction of sampling rate relative to amplifier reset time. This step alone resulted in a reduced standard deviation of baseline (negative control wells) of an order of magnitude and eliminated 4 orders of magnitude

	of baseline drift. • With support from Ames data analyst, built a matlab program for rapid data analysis and presentation of bioCD zeroday data. • In consultation with Purdue Statistics Department implemented a quantile regression statistical model to analyze ZeroDay data. o Meta analysis of ZeroDay 4-8 (prior to amplifier tuning) revealed a statistically significant response in live spores compared to dead/negative and a statistically significant effect of gravity on that response. • Achieved a clear and consistent resolution of calcium responses from germinating spores under rotation in hyper-gravity on a single bioCD. These results are the first of their kind. o After reduction of baseline noise and drift, a statistically significant, gravity dependent signal was observed in live spores (but not dead spores or empty wells) in a single bioCD experiment.
Bibliography Type:	Description: (Last Updated: 04/26/2017)
Abstracts for Journals and Proceedings	Rickus J, Park J, Roux S, Salmi M, Ricco A, Kozarsky D, Wickizer B, Rademacher A, Schooley A, Benton J, Sweet A, Tran H. "Gravity-induced Calcium Response of C. richardii during Rotation Measured by the SporeSat bioCD System." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015